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PRODUCTIVITY AND ACADEMIC POSITION IN THE SCIENTIFIC CAREER*

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This paper examines the interrelationship between scientific productivity and academic position, two key dimensions of the scientific career. Contrary to the results of most earlier studies, the effect of departmental location on productivity is found to be strong, whereas the effect of productivity on the allocation of positions is found to be weak. Productivity, as indicated by measures of publications and citations, is shown to have an insignificant effect on both the prestige of a scientist's initial academic appointment and on the outcome of institution changes later in the career. Although the relationship between productivity and the prestige of an academic appointment is insignificant at the time a position is obtained, the effect of departmental prestige on productivity increases steadily with time. For those scientists who change institutions, the prestige of the new department significantly affects changes in a scientist's productivity after the move. It is argued that past studies have obtained spurious results due to their failure to employ a longitudinal design. Not only do cross-sectional designs provide misleading results regarding the interrelationship between departmental location and productivity, but they also systematically alter the findings regarding the effects of sponsorship and doctoral training on productivity.

Two fundamental questions underlie past studies of the academic careers of scientists. First, what accounts for the unequal distribution of prestige in the academic stratification system? Why do some scientists obtain positions in highly prestigious graduate departments while others find themselves in relatively unknown departments? Second, what accounts for the great inequality in the productivity of scientists? If the institutional goal of science is the extension of certified knowledge (Merton, 1942:267-85), why do some scientists succeed, publishing widely-acclaimed work year after year, while other scientists publish almost nothing throughout their careers? Natural ability, dedication, quality of education, and sponsorship have all been considered as explanations of this unequal distribution of scientific productivity and aca-

ademic position. In addition, location in the stratification system and productivity have each been considered as dependent upon the other.

While productivity has been examined as a primary causal influence on the allocation of positions in the scientific community, and departmental location has been considered as a factor explaining the inequality in productivity, the interdependence of these two factors over the course of the scientific career has not been systematically explored. This is an important limitation since if these factors do have a reciprocal relationship, then an empirical analysis of either factor which fails to control for the other must yield biased results. This bias will not only affect conclusions about the nature of the relationship between productivity and academic position, but will also alter findings on the effects of other variables upon productivity and position.

In this paper the relationship between productivity and position is examined by studying the early careers of Ph.D. biochemists employed in U.S. graduate departments. By employing a longitudinal design, the effect of productivity on the allocation of position can be isolated from

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the effect of departmental location on productivity. We find that while the effect of productivity on the prestige of a position is weak, the effect of departmental location on productivity is strong. These findings are contrary to much recent empirical work in this area; we suggest that the failure of past studies to utilize a longitudinal design accounts for this disagreement.

PRIOR RESULTS

Prior studies have consistently reported a moderate positive relationship between scientific productivity and departmental prestige (Cole and Cole, 1973; Crane, 1965; Hagstrom, 1968; Hargens and Hagstrom, 1967; Manis, 1951). This is true even though these studies differed in research design, examined a variety of fields, and employed different measures of scientific productivity (based both on publications and citations to publications) and prestige (based on both departmental and institutional location).

Two distinct explanations of the correlation between position and productivity have been offered. First, it has been argued that more prestigious departments *select* more productive scientists for their faculty; productivity is causally prior to position. Alternatively, it has been argued that being at a more prestigious department *facilitates* greater productivity; position is causally prior to productivity. In the remainder of this section we will review these alternative explanations.

Productivity and Departmental Selection

If prestigious departments select more productive scientists for faculty positions, a correlation between a scientist's productivity and the prestige of his academic position should be found. Empirical work along this line has attempted to isolate those factors which determine the prestige of a scientist's academic appointment. Historically such research falls into two groups: the first consisting of studies of the academic profession pursued in large part out of curiosity, concern or practical interest in how the academic marketplace

is functioning; the second motivated by the work of Robert Merton on the normative structure of science, where the recruitment process is examined as an instance of the allocation of rewards by the scientific community. Researchers in the first group have consistently seen little effect of productivity on position, while researchers in the second group have, to varying extents, found a positive effect.

The best example of the first set of studies is Caplow and McGee's (1958) *The Academic Marketplace*. After examining the process by which a number of academic vacancies were filled, they concluded that hiring decisions were not based on actual evaluation of the applicant's work, but rather were based on the prestige of the candidate's graduate department, the eminence of his sponsors, and chance. Logan Wilson (1942:29, 40, 50) similarly noted the importance of the prestige that the young graduate obtains from his affiliation with his doctoral institution. And Max Weber (1919:132) warns students that chance looms large in the careers of scientists; achievement does not necessarily lead to deserved position. Thus, studies of the academic profession consistently emphasize the importance of the scientist's perceived prestige, prestige based more on the established reputations of his sponsors and doctoral department than on his actual scientific work.

The second group of studies, dealing with the social functioning of the science system rather than the characteristics of the academic profession, are founded on Merton's (1973) seminal work on the normative structure of the social system of science. It is argued that science is governed by the norm of universalism, according to which careers should be open to talent and not be determined by functionally irrelevant characteristics. Since the production of certified knowledge is the institutionally sanctioned goal of science, rewards should be allocated on the basis of an individual's contribution to the body of scientific knowledge. Appointment to a prestigious position in the hierarchy of academic departments, which is considered to be a form of recognition (Cole and Cole, 1973:57), should be

allocated to reward a scientist's productivity.

Several studies have examined this hypothesis. Cole and Cole (1973:98) found that a scientist's productivity has as strong an effect on academic appointment as does the ranking of the scientist's doctoral department. Similarly, Hargens and Hagstrom (1967) found that the effects of doctoral origins and productivity on current academic affiliation are approximately equal for their entire sample; the effect of doctoral origins is stronger for younger scientists, however. Crane (1970), on the other hand, concluded that the prestige of the doctorate has more influence than productivity upon selection for a prestigious position. She suggested (1970:961) that "despite the system's normative commitment to universalistic criteria, they are not utilized in practice."

Departmental Effects on Productivity

Less frequently the relationship between productivity and position has been considered as the result of the effects of departmental location on scientific productivity. A variety of mechanisms for a departmental effect have been proposed (cf. Crane, 1970; Hagstrom, 1968): prestigious departments provide a scientist with more free time for research, superior physical resources, abler research assistants, more stimulating colleagues, and stronger social support for doing research; and there may be a "halo" effect for being located at a prestigious department whereby papers and grant proposals appear more impressive to reviewers and to potential citers.

Crane (1965) concluded that the effects of departmental position on productivity are positive, but smaller than those of doctoral origin. Unfortunately, in her study no attempt was made to determine if the relative strengths of these effects change (with departmental effects becoming stronger) over the course of the scientist's career. Hagstrom (1967:61) concluded that while departmental prestige is associated with a number of factors which would be expected to influence productivity, there is no support for "believing that the greater productivity of scientists in high prestige departments is due more

to the context in which they work, the facilities and assistants that are available, than to their research skills and motivations."

Overall, then, no consensus emerges from past studies. Little evidence is found for a departmental effect, although tests of such an effect are relatively rare. And while the support for the importance of productivity in the selection process is stronger, substantial conflicting evidence also exists.

METHODOLOGICAL NOTE

In evaluating these studies it is important to realize that they are all limited by their designs—none is longitudinal, monitoring the work and publication histories of scientists over an extended period of time. As a result, the causal ordering of productivity and position is inherently ambiguous. Studies which explain position in terms of productivity would have been equally justified on empirical grounds to turn their analysis around and to have explained productivity in terms of position. A longitudinal research design is necessary to distinguish between the competing hypotheses of a departmental effect vs. a selection effect. In the absence of such a design, estimates of the effect of productivity on position will be upwardly biased to the extent that a departmental effect is operating, a point made by Hargens and Hagstrom (1967:30).

The following analysis employs a longitudinal design in order to untangle the interrelationships between productivity and position. The selection process is examined in two ways. First, the movement of a scientist into his first academic job from either graduate school or a postdoctoral fellowship is analyzed as a function of characteristics of his education and his productivity up to the time of his appointment. Second, the movement of a scientist from one institution to another is studied. The relative strengths of the effects of educational experiences, productivity in the period immediately prior to the move, and the prestige of the first position are assessed.

The effect of departmental location on productivity also is tested in two ways.

First the change in the productivity of a scientist after he obtains his first position is related to the prestige of his location. By examining the changes in productivity levels of those scientists who do *not* change positions, the possibility of a selection process causing the change in correlations between productivity and position can be eliminated. Second, changes in the productivity of those scientists who do change institutions are analyzed. If departmental location is a factor affecting an individual's scientific productivity, a change in productivity should be observed after departmental mobility and this change should be related to the prestige of the new department. We conclude by summarizing our findings and relating them to previous research.

DATA AND MEASUREMENT

Analysis is based on the population of male biochemists who obtained their doctorates in the fiscal years 1957, 1958, 1962 and 1963.¹ From this population of Ph.D.s two groups are isolated. The first and larger group consists of 134 biochemists who held teaching positions in graduate departments evaluated by the Roose and Andersen (1970) study and who were classified as *nonmovers*. To be considered a nonmover, a biochemist had to have held a position in the same department from the time he entered the academic marketplace (which may have occurred after a postdoctoral fellowship) until our time series ends (from ten to fifteen years after the doctorate, depending upon the year of its receipt). In order to insure adequate time series for uncovering departmental effects, the set of nonmovers also was limited to those for whom six consecutive years of data were available. The second group (N=47) consists of biochemists who held teaching positions in two different departments in two different institutions during the first ten to fifteen years of their careers. To be included

in this set of *movers*, a biochemist had to have held his position in the first location for at least three years and his position in the second location for at least four years before our time series ended. Those scientists who met these conditions but who left their teaching positions for nonteaching activities (e.g., university administration, research in industry or government) were excluded since earlier analysis (Long, 1977a:124-6) suggests that the productivity patterns of such scientists prior to their moves out of academia are atypical.

Our measure of departmental prestige is based on the Roose and Andersen (1970) study.² Since biochemists obtain teaching positions in a variety of departments other than biochemistry (e.g., chemistry, physiology) and accurate information on the actual department in which an individual was located is often impossible to obtain, an index of the overall prestige of those departments in which a biochemist might be teaching was constructed. This bioscience prestige index was defined as the weighted sum of the prestige scores of an institution's departments of biochemistry (1/12), chemistry (1/4), physiology (1/12), microbiology/bacteriology (1/12), and pharmacology (1/12), where the weights (contained in parentheses) were based on estimates of the number of biochemists teaching in each field. These scores range from 100 to 500.

Productivity was measured using counts of both publications and citations to them. *Chemical Abstracts* (1955-1973) was used to locate the articles by the biochemists in our sample. Citations to these articles, as found in the *Science Citation Index* (Institute for Scientific Information, Vols. 1961, 1964, 1966, 1968, 1970, 1972, and 1974), were used for constructing the citation measures. The name of the first author on multiple-authored papers where the cohort member was not the first author was used to locate citations to junior-authored papers; thus downward bias in counts for scientists who were predominantly junior authors was avoided (see Reskin, 1977:494-5, for

¹ Analysis was restricted to male biochemists due to the small number of females who obtained doctorates in biochemistry during this period, and to the difficulty in obtaining complete information on those females who did obtain degrees.

² These data were kindly provided by Charles J. Andersen.

details of this problem). From these publication and citation counts, measures were constructed for each year the biochemists were located in rated departments. For a given year, the publication measures reflect three years of publications and the citation measures are restricted to citations to publications in this three-year period. While in most studies citation measures cover all past publications this modification was introduced to make the measure of citations more sensitive to changes in the utilization of an individual's work. Since coverage by *Science Citation Index* and *Chemical Abstracts* increased during the period covered by our study, counts were standardized within years of the Ph.D. After standardization the scale of the measures was adjusted back to the scale of the raw counts and the minimum value was reestablished as zero. For purposes of analysis, the measures were transformed by the square root to compress the upper end of the original metric. Exact details on the construction of yearly citation measures are presented in the appendix.

Finally, a series of measures to reflect the biochemists' educational experiences were coded. Since approximately two-thirds of the biochemists held postdoctoral fellowships,³ the prestige of the postdoctoral institution was coded using the bioscience index described above. A measure of the prestige of the last educational location was constructed by assigning a biochemist the prestige score of his doctoral department if he did not have a postdoctoral fellowship in a graduate department, and the prestige of the fellowship institution if he did. This was done since earlier analysis (Long, 1977b) indicates that for postdoctoral fellows the prestige of the postdoctoral location replaces the prestige of the doctoral department in affecting career outcomes.

A measure of the mentor's accom-

plishments proximate to the period when the sample members obtained their degrees was constructed by counting citations to the mentor's articles published in the five years prior to 1961. This measure will be referred to variously as the "mentor's eminence," the "mentor's prestige," or the "recognition received by the mentor." It should be kept in mind, however, that citations are not an unambiguous indicator, since they may reflect both the performance of a scientist and his or her standing or prestige in the scientific community. Distinguishing between these two dimensions of a mentor's possible influences on the careers of his students is an important topic, but one that is beyond the scope of this paper (cf. Reskin, 1978).

Collaboration with the mentor was indicated by a dummy variable assigned a value of one if the mentor coauthored on one or more articles published before or during the year after the student's receipt of the doctorate, and zero otherwise. The year after the Ph.D. was included in the determination of predoctoral collaboration to reflect the lag between the completion of research and its publication.

Astin's (1971) score for the selectivity of an institution was assigned to each baccalaureate institution. The selectivity index has values ranging from one to seven, with seven being the most selective category. This variable, which has been interpreted as a crude indicator of intelligence and as a measure of the quality of baccalaureate education, has been shown to be a moderately successful predictor of future success.

PRODUCTIVITY AND THE ALLOCATION OF ACADEMIC POSITIONS

To assess the magnitude of the effect of productivity on appointment to a prestigious position, productivity must be measured at the time a scientist obtains that position, not several years later. For if productivity measures are based on work completed in the department of the appointment, the effect of departmental location on productivity may bias the estimates. This is demonstrated in Table 1, which presents the results of regressions on the prestige of the initial academic ap-

³ While two-thirds of the biochemists studied in this paper began their careers with postdoctoral fellowships, it should be understood that among the entire cohort (i.e., both those with academic and nonacademic careers) the fraction having immediate postdoctoral fellowships is one-half. This fraction is in agreement with estimates from other sources (cf. Curtis, 1969).

Table 1. Regressions Relating Predoctoral Statuses and Productivity to Prestige of First Academic Position, for Those 134 Biochemists Who Do Not Change Institutions

Equation		Coefficients ^a of					R ²	d.f.
		ORIG	MENT	SEL	PUB	CIT		
1. Prestige Regressed on Productivity at Job Year 1	β	.370	.202	.145	-.101	.152	.304	127
	t	4.66	2.48	1.89	1.01	1.49		
	r	.464	.372	.249	.096	.204		
2. Prestige Regressed on Productivity at Job Year 7	β	.295	.173	.128	.015	.270	.358	127
	t	3.72	2.23	1.75	0.16	2.78		
	r	.464	.372	.249	.327	.446		

NOTE: Dependent variable is the Roose-Andersen bioscience prestige score of the institutional location. Item identifications are: ORIG=origin prestige, Cartter prestige of Ph.D. department if non-fellow or if nonacademic fellow and Roose-Andersen bioscience prestige score if fellow; MENT=square root of five year citations counts to mentor; SEL=selectivity of baccalaureate institution; PUB=publication level, square roots of standardized levels of three-year publication counts ending in the given year; CIT=citation level, square roots of standardized levels of citations to publications in the three-year period ending in the given year.

^a Row β gives standardized regression coefficients; row t gives the t-statistics (with more than 120 degrees of freedom in the regression, critical values for a two-tailed test of significance at the .10, .05 and .01 levels are: 1.645, 1.960 and 2.576, respectively; for a one-tailed test the critical values are: 1.282, 1.645 and 2.326 for significance levels of .10, .05 and .01, respectively); row r gives zero-order correlation coefficients with dependent variable.

pointments for the 134 biochemists described above as nonmovers.⁴

Equation 1 regresses departmental prestige on characteristics of the scientist's training and productivity. The productivity measures are based only on work completed before the scientist obtained his first position.⁵ Thus the correlations between productivity and position, .096 for publications and .204 for citations, cannot be the result of a departmental effect. These correlations are substantially smaller than those reported in the studies reviewed above.⁶ Further, the standardized regression coefficients from publications and citations are small and statistically insignificant, while those from the prestige of the sponsoring department

(either the doctoral or fellowship department), the mentor's eminence, and the selectivity of the baccalaureate institution are strong, positive and statistically significant. Thus no support is found for the importance of productivity in the recruitment process.⁷

The importance of using productivity measures based only on work completed at the time of the appointment can be seen in equation 2 of Table 1. This equation includes productivity measures based on work completed in the three-year period ending in the sixth year of the scientist's location at the department whose prestige is the dependent variable. The standardized regression coefficient from citations to the prestige of the academic appointment has increased substantially from .152 to .270. The coefficient is now

⁴ The process by which biochemists obtain their initial academic positions has been examined in detail elsewhere (Long, 1977b). Here we only report those results which have the greatest impact on our argument concerning the relative importance of selection and departmental effects.

⁵ To take into account the time lag between the completion of research and its publication in a professional journal, we included research published during the scientist's first year at his new position in these productivity measures.

⁶ Crane (1970) is an exception which will be considered below.

⁷ Scientists who held their first position in the department where they obtained their doctorate or pursued postdoctoral studies (i.e., inbred scientists) have been included in these and later analyses. When these scientists are excluded, the coefficient from the origin prestige is attenuated but the other coefficients remain essentially the same. Substantive conclusions are not affected by the inclusion of inbred scientists in the reported results (cf. Hagstrom and Hargens, 1968).

the largest in the equation and it has attained statistical significance. Further, the magnitudes of the coefficients from the prestige of the doctoral origin and the citation level are now approximately equal to those reported by Cole and Cole (1973:98) for a similar regression.⁸ But clearly equation 2 is not the appropriate test of the selection hypothesis since it "explains" departmental prestige on the basis of work completed at that department, and thus confounds the selection effect with the departmental effect. Yet this is essentially what is being done in cross-sectional studies that relate the prestige of a scientist's location at a given time to his productivity at that time, not at the time he obtained the position.

Note also that the apparent increase in the effect of productivity is associated with an attenuation of the effects of other variables, even though the zero-order correlations for these variables are necessarily unchanged. Thus the regression effect of the prestige of the sponsoring department decreases from .370 to .295, the effect of the mentor's prestige decreases from .202 to .173, and the effect of the baccalaureate institution decreases from .145 to .128. So, not only is the productivity effect overestimated in a cross-sectional design, but the effects of other variables are underestimated. In short, changing from a cross-sectional to a longitudinal design results in substantively and theoretically different conclusions.

Before rejecting the operation of a selection effect, a further complication must be considered. It is plausible that departments hiring young scientists do not base their selection on the demonstrated scientific contributions of the candidates since the young scientists have not had adequate time to establish their scientific

merit. Professionally older scientists, who have had the time to publish a body of research, may be evaluated and recruited on the basis of their published work. Hargens and Hagstrom (1967:34) suggest that in this way the stratification system reduces the initial inequalities in the allocation of position.

Our data do not support this argument. First, in an analysis not reported here (see Long, 1977b) it is found that for biochemists who begin their careers with postdoctoral fellowships the delayed process of job allocation does not differ significantly from the allocation process for those obtaining positions immediately after receiving the doctorate. Since the fellows have devoted several additional years to research prior to entering the job market,⁹ one would expect on the basis of the above hypothesis that their recruitment would be based more closely on their productivity.

Second, the institution changes which occur later in the career do not seem to be affected by productivity. Table 2 presents regressions on the outcomes of 47 institution changes occurring on average 7.6 years after the doctorate.¹⁰ In equation 1 the destination prestige of the move is regressed on the prestige of the scientist's initial academic department, characteristics of his training, and his productivity at the time of the move. The effect of citations on the prestige of the academic appointment has increased slightly from equation 1 of Table 1, and with more observations we would expect the coefficient to become statistically significant. Still, the magnitude of the effect remains small and does not offer strong support for the hypothesis of the universalistic allocation of positions later in the career. Indeed, the scientist's productivity at the

⁸ It is generally argued that only unstandardized regression coefficients should be compared across populations. But, as Hargens (1976) has argued, in cases where it is theoretically reasonable to expect standardized coefficients to be structural parameters, the comparison of standardized coefficients is justified. Certainly in the case of evaluating a scientist's productivity it is not his number of publications or citations that is important, but rather his number relative to other scientists in his field. Accordingly, the comparison of standardized coefficients is appropriate in this case.

⁹ On average the postdoctoral fellows in our sample began their first tenure-track faculty position 3.2 years after the doctorate, with 60% of the fellows having positions lasting three or more years.

¹⁰ While the results on institution changes are limited by the small number of cases, we note that similar findings were obtained in analyses of 72 institution changes (which include these 47 cases). The results presented here are limited to 47 cases since only these cases provide sufficient time series for testing the impact of departmental location on productivity (analysis presented below).

Table 2. Regressions Relating Predoctoral Statuses, Productivity and Prestige of First Institutional Location to Prestige of Second Institutional Location for Those 47 Biochemists Who Held a Position at the First Institution for at Least Three Years and at the Second Institution for at Least Four Years

Equation		Coefficients ^a of						R ²	d.f.
		PRST	ORIG	MENT	SEL	PUB	CIT		
1. Prestige									
Regressed on									
Productivity at	β	-.173	.081	.121	.105	-.111	.195	.069	40
the Time of the	t	0.98	0.48	0.75	0.66	0.52	0.94		
Institution Change	r	-.088	.093	.124	.102	-.001	.102		
2. Prestige									
Regressed on									
Productivity Three	β	-.175	.002	.076	.027	-.012	.387	.167	40
Years after the	t	1.04	0.00	0.49	0.18	0.05	1.86		
Institution Change	r	-.088	.094	.124	.102	.195	.365		

NOTE: Dependent variable is the Roose-Andersen bioscience prestige score of the second institutional location. Item identifications are: PRST=prestige of first institutional location, Roose-Andersen bioscience prestige score; ORIG=origin prestige, Cartter prestige of Ph.D. department if nonfellow or if nonacademic fellow and Roose-Andersen bioscience prestige score if fellow; MENT=square root of five-year citation counts for mentor; SEL=selectivity of baccalaureate institution; PUB=publication level, square roots of standardized levels of three-year publication counts ending in the given year; CIT=citation level, square roots of standardized values of citations to publications in the three-year period ending in the given year.

^a Row β gives standardized regression coefficients; row t gives the t-statistics (with 40 degrees of freedom in the regression, critical values for a two-tailed test of significance at the .10, .05 and .01 levels are: 1.684, 2.021 and 2.704 respectively; for a one-tailed test the critical values are: 1.303, 1.684 and 2.423 for significance levels of .10, .05 and .01, respectively); row r gives the zero-order correlations with the dependent variable.

time of the move is more strongly related to the prestige of the origin institution (zero-order correlations of .457 with publications and .382 with citations) than with the prestige of the destination institution (zero-order correlations of -.001 with publications and .102 with citations). Later career mobility does not operate in such a way as to create greater correspondence between the prestige of a scientist's academic affiliation and his productivity.

Somewhat surprisingly, no other variables in equation 1 of Table 2 have a significant relationship to the prestige of the destination institution. The prestige of the doctoral origin no longer aids the scientist in obtaining a position. The mentor appears to continue to affect his student's placement, but the positive effect is greatly attenuated. Finally, there is no strong relationship between the prestige of the first institution of employment and the prestige of the destination institution.

There is some evidence that the tradeoff

between academic rank and institutional prestige, and the negative effects of age influence the outcomes of institutional mobility. Caplow and McGee (1958:148) suggest that downward mobility is often accompanied by advancement in rank. For all biochemists who change institutions during the period our data covers (which includes the 47 cases in Table 2), 62% of the scientists moving to a less prestigious department advanced in academic rank, while only 37% of those moving to a more prestigious department advanced in rank. Caplow and McGee (1958:44, 186-7) also suggest that older scientists are professionally less attractive since their future productivity may be less than that of younger scientists. And indeed, we find that the older the scientist the less likely he is to move to a more prestigious institution.

Still, none of the factors affecting the outcome of institution changes has a strong effect. This suggests that the mechanism by which job changes occur is

different from that determining the initial deployment of scientists, where doctoral origins and characteristics of the mentor have a significant effect. This inability to uncover factors that strongly influence job changes may be related to Caplow and McGee's (1958:47, 53, 55, 77, 82, 91, 160) finding that a wide variety of both personal and professional factors influence a scientist's decision to change institutions. The uniqueness of these factors cannot be tapped with the variables in our study.

These findings may be compared with two similar analyses. Crane (1970) examined the recruitment of faculty members in six departments among the 20 most prestigious graduate schools. While she did not report regression coefficients predicting the destination prestige and combined those who obtained their first position with those who changed institutions, she indicated that the correlations with destination prestige and the number of publications is $-.051$ and the correlation with the number of citations is $-.001$, with these measures being based on work completed *before* the move. She also reported that productivity had no effect on the prestige of the position obtained.

Allison (1976:205-18) analyzed the institution changes of 894 scientists from Hagstrom's (1967) study of four fields. Unlike Crane's and our findings, productivity emerged as a significant factor affecting destination prestige. But, as Allison (1976:185) warned, his productivity measure is based on citations made *after* the job changes. Hence if changes in productivity which occur after the job change are systematically related to the prestige of the destination department, his coefficient for the effect of productivity will be confounded with the effect of departmental location on productivity. Results presented in equation 2 of Table 2 indicate how this may occur. In this equation productivity measures are based on work completed after the job change. Both the correlations of productivity with destination prestige and the coefficient from citations increase substantially. The utilization of the scientist's work (as indicated by citations to it) appears to influence significantly the outcome of his institution change. Thus, given Allison's warning

about his productivity measures and the findings of equation 2, we can explain the difference between his results and ours.

Overall there is no evidence of the importance of productivity in the recruitment of faculty. Rather, in the initial placement of graduates the prestige of the sponsoring department and the eminence of the mentor play major roles. Institution changes occurring later in the career appear to be unaffected by either characteristics of the education of the scientist or by his productivity. Moreover, if productivity measures which are based on work completed after appointment to a department are used to predict the prestige of that department, it will *appear* as though productivity is a causally significant factor. These findings suggest the importance of a departmental effect, an issue which is now considered.

EFFECTS OF DEPARTMENTAL LOCATION ON PRODUCTIVITY

Tables 3 through 6 present evidence for the existence of a departmental effect on productivity. Tables 3 and 4 examine the effects of departmental location on publications and citations to publications¹¹ for those 134 biochemists who do *not* change institutions during the period for which we have data. Accordingly, changes in the relationship between departmental prestige and productivity cannot be attributed to the awarding of more prestigious positions to those who are productive. Tables 5 and 6 examine the effects of departmental location on those who change institutions during the period for which we have data. This allows us to explore the effects of changes in departmental location on productivity.

For scientists who do not change departments, publication levels are not immediately affected by departmental prestige, as is seen in equations 1 and 2 of Table 3. Equation 1 shows that produc-

¹¹ Recall that the publication measure is based only on publications in a three-year period ending in the given year. Similarly, the citation measure is based only on citations to publications in this three-year period, not to all past publications. This was done to make the productivity measure more sensitive to changes over time.

Table 3. Regressions Relating Predoctoral Statuses, Institutional Location and Earlier Productivity to Publication Level, for Those 134 Biochemists Who Do Not Change Institutions

Equation		Coefficients ^a of								R ^a	d.f.
		PRST	ORIG	MENT	COL-LAB	SEL	PUB (Yr-3)	CIT (Yr-3)			
1. Publication Level at Job Year 3	β	-.038	-.005	.239	.135	.034			.084	127	
	t	0.38	0.05	2.50	1.55	0.39					
2. Publication Level at Job Year 3	β	-.046	-.026	.184	.047	.005	.211	.153	.179	125	
	t	0.48	0.27	1.99	0.54	0.05	1.83	1.32			
	r	.066	.078	.255	.177	.055	.359	.348			
3. Publication Level at Job Year 6	β	.220	.181	-.039	.171	.103			.170	127	
	t	2.29	1.93	0.43	2.06	1.22					
4. Publication Level at Job Year 6	β	.252	.180	-.169	.095	.087	.593	-.049	.463	125	
	t	3.14	2.37	2.22	1.40	1.27	6.82	0.54			
	r	.327	.300	.157	.192	.166	.572	.393			

NOTE: Dependent variables are square roots of standardized three-year publication counts ending in the given year. Item identifications are: PRST=prestige of institutional location, Roose-Andersen bioscience prestige score; ORIG=origin prestige, Cartter prestige of Ph.D. department if nonfellow or if nonacademic fellow and Roose-Andersen bioscience prestige score if fellow; MENT=square root of five-year citation counts for mentor; COLLAB=collaboration with mentor, defined as one if any article published by the year after the doctorate was coauthored with the mentor, zero otherwise; SEL=selectivity of baccalaureate institution; PUB(Yr-3)=publication level three years earlier, square roots of standardized levels of three-year publication counts ending three years before the given year; CIT(Yr-3)=citation level three years earlier, square roots of standardized levels of citations to publications in the three-year period ending three years before the given year.

^a Row β gives standardized regression coefficients; row t gives the t-statistics; row r gives zero-order correlations with dependent variable.

tivity levels are affected by the prestige of the scientist's mentor during his doctoral study and whether or not he collaborated with his mentor. By controlling for earlier productivity, in equation 2, insights into how the mentor influences his student's later productivity may be obtained. Later in the career collaboration does not directly affect productivity, but rather operates through its effects on the early productivity of the student. For at the time of doctoral studies, collaboration with the mentor accounts for a substantial proportion of the student's publications. Even though actual collaboration with the mentor usually ceases after receipt of the degree, the influence of early publications continues to affect the student's rate of publication. The strong positive effect of early publications on later publications indicates a tendency of those who publish to continue publishing. There remains, however, a good deal of variation over time in individual levels of productivity as indicated by the correlation of only .359 between publication levels in consecutive

three-year periods. While the effect on publications of the mentor's prestige remains significant, it is somewhat attenuated by the introduction of controls for early productivity. This, as in the case of collaboration, results from a positive, albeit weaker, effect of the mentor's prestige on the student's predoctoral productivity.

Equations 3 and 4 show that departmental prestige significantly influences the scientist's productivity after the third year in a department. In equation 3 we find that departmental location is the strongest factor explaining the scientist's level of publications. Note that this effect persists after controlling for earlier productivity, as seen in equation 4. Thus *change* in the publication rate is most strongly affected by the prestige of the scientist's academic location—scientists in prestigious departments increase their publication rate while those in less prestigious departments begin to publish relatively less.

As departmental effects emerge, the effects of variables associated with the

biochemist's doctoral education change. The effects of the prestige of the doctoral origin on the level of publications and on the change in the publication rate increase substantially. The total effect operates in two ways. First, the prestige of the doctorate acts indirectly through its effect on the prestige of the scientist's academic appointment, which in turn affects the scientist's publication rate. Second, the prestige of the doctorate directly affects the biochemist's later publication pattern. This direct effect would be consistent with the argument that students at better institutions are more talented or better trained, and that in time this talent or training emerges through greater productivity. An alternative hypothesis, however, is that the greater number of informal contacts which students in prestigious departments have the opportunity to make eventually pays off in resources that facilitate productivity. While the effect of collaboration with the mentor on the level of publications increases, this remains an

indirect effect operating through the effect of early productivity on later productivity, as was the case in equations 1 and 2. Finally, the effect of the mentor's eminence on changes in the student's rate of publication later in the career has become *negative*, of moderate strength, and statistically significant. This is an intriguing and unanticipated finding which indicates that after a scientist's productivity pattern has been established (in this case, six years into the teaching career), those scientists with more prestigious mentors begin to publish less. The fact that the effect of the mentor's eminence on the citations received by a scientist is insignificant (Table 4, equation 4), suggests that scientists with more eminent mentors begin to be more selective in their publishing practices, publishing less frequently, *ceteris paribus*, than other scientists, but maintaining their total number of citations.

Results for citation levels are presented in Table 4. As with publications, the departmental effect is strong and positive.

Table 4. Regressions Relating Predoctoral Statuses, Institutional Location and Earlier Productivity to Citation Level, for Those 134 Biochemists Who Do Not Change Institutions

Equation		Coefficients * of							R ²	d.f.	
		PRST	ORIG	MENT	COL-LAB	SEL	PUB (Yr-3)	CIT (Yr-3)			
1.	Citation Level at Job Year 3	β	.192	-.068	.239	.074	.097			.147	127
		t	1.97	0.71	2.59	0.89	1.14				
2.	Citation Level at Job Year 3	β	.178	-.080	.164	-.003	.040	.056	.343	.273	125
		t	1.96	0.90	1.87	0.03	0.49	0.51	3.14		
		r	.279	.125	.317	.124	.170	.332	.455		
3.	Citation Level at Job Year 6	β	.335	.123	-.063	.122	.087			.243	127
		t	3.65	1.37	0.73	1.54	1.09				
4.	Citation Level at Job Year 6	β	.280	.147	-.080	.061	.045	.264	.338	.501	125
		t	3.62	2.00	1.10	0.94	0.69	3.15	3.89		
		r	.445	.324	.269	.167	.190	.500	.591		

NOTE: Dependent variables are square roots of standardized levels of citations to publications in the three-year period ending in the given year. Item identifications are: PRST=prestige of institutional location, Roose-Andersen bioscience prestige score; ORIG=origin prestige, Cartter prestige of Ph.D. department if nonfellow or if nonacademic fellow and Roose-Andersen bioscience prestige score if fellow; MENT=square root of five-year citation counts for mentor; COLLAB=collaboration with mentor, defined as one if any article published by the year after the doctorate was co-authored with the mentor, zero otherwise; SEL=selectivity of baccalaureate institution; PUB(Yr-3)=publication level three years earlier, square roots of standardized levels of three year publication counts ending three years before the given year; CIT(Yr-3)=citation level three years earlier, square roots of standardized levels of citations to publications in the three-year period ending three years before the given year.

* Row β gives standardized regression coefficients; row t gives t-statistics; row r gives zero-order correlations with dependent variable.

The effect of the scientist's departmental prestige on the utilization of his work (as indicated by citations to it), begins more rapidly after the doctorate and is of greater magnitude than was the case with publications. The citations received by work completed in the first three years at the department are significantly affected by the prestige of the department (equation 1); this effect is maintained after controlling for earlier productivity (equation 2). The departmental effect increases so that by the sixth year at the department the effect on the level of citations (equation 3) is .335, the strongest coefficient in the regression, and the effect of departmental prestige on the change in citations received (equation 4) has increased to .280 and is only slightly less than the effect of early citations on later citations. We may conclude that as a group scientists at prestigious departments produce more highly cited work, regardless of the quality of their education, the prestige of their sponsors, or their earlier productivity.

As with publications, the effects of the mentor are largely indirect after the scientist obtains his doctorate. The correlations between the recognition received by the mentor and the student's citation level decrease with time. By controlling for earlier productivity the effect is shown to be indirect, as was the case with publications. By year six the direct effect has been completely eliminated. Similarly, the effect of collaboration operates indirectly through the impact of collaboration on early publications and citations, and through the effects of this earlier productivity on later productivity. Thus, equations 2 and 4 show that while the correlations with collaboration are in fact increasing, the direct effects remain small. The effect of doctoral prestige on citations increases with time and persists after controlling for earlier productivity, as was the case with publications.

Finally, while there is somewhat more stability in citations than in publications, a great deal of variation over time persists. The correlation between the citations received in the three-year periods ending in years four and seven is only .591. This is less than the figures reported in other

analyses (cf. Cole and Cole, 1973:32). The smaller stability is probably due to our construction of citation measures which reflect the utilization of work *published* in different time periods; thus the effect of the same papers receiving citations in several consecutive years was eliminated. These results are interesting in and of themselves, indicating that a great deal of variability exists in the number of citations a scientist's work receives over time, even though the total number of citations (i.e., citations to all past work) he receives in a given year may be quite stable.

So far our results have shown that controlling for training and earlier productivity, departmental location has a substantial effect on the quantity and utilization (as indicated by citations) of a scientist's work. It is of course possible that the increasing correspondence between productivity levels and the prestige of a scientist's departmental location reflects the hiring institution's ability to judge a scientist's merit by means other than his published work. Thus, if a scientist has only a dissertation and a few publications, or perhaps none at all, at the time he applies for a job, it may be that less formal indicators are evaluated by the hiring department and the departmental effect we uncovered above merely reflects the validation of the department's initial judgments.

To test further our hypothesis of a departmental effect, we examine the productivity of scientists who change institutions.¹² If environment does influence productivity, we would expect the institution changes of scientists to be followed by changes in their productivity which reflect their new environments. Tables 5 and 6 present the results of regressions on productivity levels of 47 scientists who change departments.

Equations 1 and 2 of Table 5 show that while at the first employing (origin) department the effect of departmental location on the scientist's publication level continues to increase until the time of the move. In equation 3 we see that this effect persists after controlling for earlier pro-

¹² This approach was developed in collaboration with Paul Allison. Together we plan a more detailed study of a much larger sample of job changes.

Table 5. Regressions Relating Predoctoral Statuses, Institutional Locations and Prior Productivity to Publication Levels, for Those 47 Biochemists Who Held a Position at the First Institution for at Least Three Years and at the Second Institution for at Least Four Years

Equation	Coefficients ^a of										R ²	d.f.
	PRST1	PRST2	ORIG	MENT	COLLAB	SEL	PUB (Yr-3)	CIT (Yr-3)				
1. Publication Level Two Years Before Move	β t r	.124 0.87 .265		.423 2.95 .414	-.123 0.86 .088	.322 2.30 .276	.019 0.14 .105			.288	41	
2. Publication Level First Year at Second Job	β t	.363 2.53		.211 1.46	-.017 0.11	.126 0.89	.101 0.73			.271	41	
3. Publication Level First Year at Second Job	β t r	.301 2.09 .457		.079 0.50 .314	.053 0.36 .124	.072 0.49 .154	.084 0.62 .218	.025 0.11 .391	.288 1.33 .454	.339	39	
4. Publication Level Fourth Year at Second Job	β t	.362 2.49	.206 1.50	.069 0.48	.095 0.65	.233 1.64	-.000 0.00			.286	40	
5. Publication Level Fourth Year at Second Job	β t r	.103 0.88 .392	.184 1.78 .195	-.096 0.85 .206	.134 1.21 .243	.143 1.34 .288	-.075 0.71 .162	.559 4.01 .719	.185 1.36 .572	.626	38	

NOTE: Dependent variables are square roots of standardized three-year publication counts ending in the given year. Item identifications are PRST1=prestige of first institutional location, Roose-Andersen bioscience prestige score; PRST2=prestige of second institutional location, Roose-Andersen bioscience prestige score; ORIG=origin prestige, Cartier prestige of Ph.D. department if nonfellow or if nonacademic fellow and Roose-Andersen bioscience prestige score if fellow; MENT=square root of five-year citation counts for mentor; COLLAB=collaboration with mentor, defined as one if any article published by the year after the doctorate was coauthored with the mentor, zero otherwise; SEL=selectivity of baccalaureate institution; PUB(Yr-3)=publication level three years earlier, square roots of standardized levels of citations to publication in the three-year period ending three years before the given year; CIT(Yr-3)=citation level three years earlier, square roots of standardized levels of citations to publication in the three-year period ending three years before the given year.

^a Row β gives standardized regression coefficients; row t gives t-statistics; row r gives zero-order correlations with dependent variable.

ductivity. Thus, while those scientists who change institutions are located at the first institution, the effects of departmental prestige on publications operates in much the same way as it did for those scientists who did not change institutions.

At the time of the move the correlation between publications and the destination prestige is $-.001$, while the correlation with the origin prestige is $.457$. By the fourth year at the new institution, the effect of the old department on the publication level at this time is greater than the effect of the new department when prior productivity is not being controlled (equation 4). But when controls for earlier productivity are introduced (equation 5), the effect of the origin department becomes greatly attenuated and is no longer statistically significant. The effect of the destination prestige, however, decreases only slightly and becomes significant at the $.05$ level. Thus, the *change* in the publication level which occurs after departmental mobility is more closely related to the prestige of the new department than to that of the old department.

Similar findings are presented for citations in Table 6. Comparison of equations 1 and 2 shows that once more the effects of departmental location on citation levels increase before the move. Also, the effect of departmental location is maintained after controlling for earlier productivity. At the time of the move the correlation between the origin prestige and the levels of citations is $.382$, while the correlations with the destination prestige is $.102$. By the fourth year at the new location, however, the effect of the old department's prestige has decreased substantially to $.102$, while the effect of the new department's prestige is $.329$. Further, by comparing equations 4 and 5 we see that the effect of the origin department on later productivity is completely eliminated by controlling for earlier productivity, while the effect of the new departmental environment stays strong and statistically significant.

SUMMARY AND CONCLUSIONS

Our findings may be summarized briefly as follows:

1. Neither the number of publications a biochemist has produced nor the number of citations he has received significantly affects the prestige of his initial academic appointment. Rather, factors related to his graduate education, sponsorship, and postdoctoral study have the greatest influence on his initial academic placement.

2. While there is only a small correlation between productivity and position at the start of a biochemist's employment in an academic department, a stronger relationship emerges through time. This departmental effect occurs independently of the reinforcing effects of earlier productivity on current productivity. Departmental prestige has a much stronger and more immediate effect on the citations a scientist's subsequent work receives than on the quantity of his publications.

3. Later in the career when departmental mobility occurs, the effects of productivity on the prestige of the new employing department increase slightly, but remain quite weak.

4. For scientists who change institutions, the effects of the first department's prestige on productivity increase until the move occurs, at which point they begin to decrease. At the same time, while the correlation between productivity and the new department's prestige is negligible when the position is obtained, the correlations increase markedly within five years. Further, the changes in productivity levels occurring after the move appear to be directly related to the new department's prestige and unrelated to the old department's prestige.

Thus, our longitudinal analysis provides consistent support for the hypothesis that productivity is facilitated by departmental location. Contrary to earlier findings, productivity, as indicated by measures of publication and citation, plays an insignificant role in the selection process. The differences between our results and those of earlier studies may be explained by the failure of past studies to employ longitudinal controls.

Our findings on the placement process

Table 6. Regressions Relating Predoctoral Statuses, Institutional Locations and Prior Productivity to Citation Levels, for Those 47 Biochemists Who Held a Position at the First Institution for at Least Three Years and at the Second Institution for at Least Four Years

Equation		Coefficients* of							R ²	d.f.
		PRST1	PRST2	ORIG	MENT	COLLAB	SEL	PUB (Yr-3)		
1. Citation Level Two Years Before Move	β	.204		.422	-.232	.160	.058		.280	41
	t	1.43		2.93	1.61	1.13	0.41			
	r	.330	-.129	.422	-.054	.094	.121			
2. Citation Level First Year at Second Job	β	.282		.270	-.149	.103	.114		.226	41
	t	1.91		1.81	0.99	0.71	0.80			
3. Citation Level First Year at Second Job	β	.256		.197	-.120	.058	.109	.108	.244	39
	t	1.67		1.17	0.77	0.37	0.75	0.46		
	r	.382	.102	.320	-.006	.085	.191	.324		
4. Citation Level Fourth Year at Second Job	β	.102	.329	.255	.006	.066	.188		.283	40
	t	0.70	2.40	1.75	0.04	0.46	1.34			
5. Citation Level Fourth Year at Second Job	β	-.062	.296	.136	.056	.008	.138	.215	.430	38
	t	0.43	2.34	0.98	0.41	0.06	1.06	1.25		
	r	.198	.365	.327	.168	.095	.274	.453		

NOTE: Dependent variables are square roots of standardized levels of citations to publications in the three-year period ending in the given year. Item identifications are: PRST1=prestige of first institutional location, Roose-Andersen bioscience prestige score; PRST2=prestige of second institutional location, Roose-Andersen bioscience prestige score; ORIG=origin prestige, Cartier prestige of Ph.D. department if nonfellow or if nonacademic fellow and Roose-Andersen bioscience prestige score if fellow; MENT=square root of five-year citation counts for mentor; COLLAB=collaboration with mentor, defined as one if any article published by the year after the doctorate was coauthored with the mentor, zero otherwise; SEL=selectivity of baccalaureate institution; PUB(Yr-3)=publication level three years earlier, square roots of standardized levels of three-year publication counts ending three years before the given year; CIT(Yr-3)=citation level three years earlier, square roots of standardized levels of citations to publications in the three-year period ending three years before the given year.

* Row β gives standardized regression coefficients; row t gives t-statistics; row r gives zero-order correlations with dependent variable.

are consistent with such studies as those by Caplow and McGee (1958) and Wilson (1942). Studies by Cole and Cole (1973), Hargens and Hagstrom (1967), and Allison (1976) appear to show a positive effect of productivity on positional prestige which is approximately equal to the effect of the doctoral prestige. But this difference may be due to an important assumption these latter studies have made. Hargens and Hagstrom (1967:30) characterize it in this way:

For our purposes, we will adopt the working assumption that a scholar's productivity rate precedes his placement in the academic stratification system and is not greatly affected by his placement. Insofar as a scholar's placement does affect his productivity rate, this assumption will lead us to overestimate the effect of his productivity (and thus of his scholarly merit) in the determination of his location in the stratification system.

All of these studies have made this assumption. Further this would explain the exceedingly low correlation which Crane observed. For in her study she obtained moves occurring in academic years 1963–1965 and used bibliographic sources from the period prior to 1964. Since the majority of Crane's cases involved instances of institutional mobility, as opposed to initial placement, her correlations of prestige with productivity should be approximately equal to those in Table 4, which is in fact the case. Our findings of departmental effects also would be consistent with Hargens and Hagstrom's (1967) findings that the effects of productivity were stronger for the older scientists than the younger. They explained this as the emergence of stronger effects of productivity as a larger body of research became available for evaluation. Alternatively it may reflect the fact that the younger members of their sample had been at their current department for a shorter period of time before productivity was measured, and thus the strength of the departmental effect would be smaller. Similarly, Hagstrom and Hargens (1968) find that the effect of citations on current prestige increases in four path models dealing with steadily older cohorts. While these earlier studies, with the exception of Crane's, do not allow the precise determination of

how long the sample members were at their current departments before productivity was measured, these explanations appear convincing.

The strength of the departmental effect which we find does not contradict Hagstrom's (1967:61) conclusion that there is no basis for "believing that the greater productivity of scientists in high prestige departments is due more to the context in which they work, the facilities and assistants that are available, than to their research skills and motivations." For, as he notes, the fact that one does not find strong intervening factors which explain the importance of prestige upon productivity does not eliminate the possibility that other, unmeasured characteristics are important (Hagstrom, 1968:12). And, as Hagstrom also notes, a longitudinal study showing the emergence of the relationship between productivity and departmental affiliation is necessary. That is what we have provided.

While several important conclusions have been drawn from our analysis, our final understanding of the relationship between productivity and departmental position remains unsatisfactory. Just how does departmental prestige facilitate productivity? Several hypotheses can be constructed, although we do not have sufficient evidence with which to substantiate any of them. First, superior departments may have the prescience to select those who will become (even though they are not now) productive. This would, to some extent, salvage the hypothesis that positions are allocated on the basis of contribution—by adding the qualification, *potential* contribution. Yet it seems unlikely that when departmental mobility occurs (which as Caplow and McGee [1958] have indicated occurs for a myriad of nonprofessional reasons) members of the new department realize that the hired individual's level of productivity will change. Second, the relationship between productivity and departmental location may be due to the effects of departmental characteristics other than prestige. For example, prestigious departments tend to be larger and hence may provide more opportunities for collaboration, which may increase the scientist's overall pro-

ductivity.¹³ Prestige is only one aspect of the context in which a scientist works. It is possible that the departmental effect we have found *underestimates* the global effect of departmental location on productivity. Third, there may be a bias in the evaluation system whereby papers from more prestigious departments *appear* superior (see Zuckerman and Merton, 1971).

Each of these hypotheses will require substantial work to confirm or reject. Given the evidence for a departmental effect which we have presented, inquiry into the above hypotheses is an important next step in studying the scientific career.

Our findings have major significance along two dimensions, one methodological and one substantive. The results presented in Tables 1 and 2 may be considered as examples of what Schoenberg (1977) referred to as *dynamic misspecification*: the application of static models to dynamic processes. Our results have clearly shown that cross-sectional designs can result in biased estimates of structural parameters. In our particular example, the effects of productivity on the allocation of position are inflated, while effects of characteristics of educational experience are attenuated. This substantive example, which illustrates the mathematical conclusions of Schoenberg's work, indicates that great care must be exercised in the interpretation of cross-sectional analysis. Since this is a general problem, not simply relevant to the process of job allocation, it is possible that results of cross-sectional analyses of other dimensions of the exchange process in science may be biased due to dynamic misspecification.

Substantively our findings call for a reconsideration of operation of the reward system in the scientific career. Merton (1973:293) has described the universalistic operation of the exchange process in science as follows:

When the institution of science works efficiently—and like other social institutions, it does not always do so—recognition and esteem accrue to those who have best fulfilled their roles, to those who have made original contributions to the body of scientific knowledge.

Cole and Cole (1973:235) have argued that the major violation of the norm of universalism involves the process of accumulative advantage: "People who have done well at time 1 have a better chance of doing well at time 2, independently of their objective role-performance; the initially successful are given advantage in subsequent competition for rewards." Our results suggest that those who receive prestigious positions at time 1 have a better chance of doing well (in terms of productivity) at time 2, independently of their earlier productivity. But this process differs in one important respect from the process described by Cole and Cole. Advantage accumulates not to those who have been successful at time 1, but to those who have received the advantage of a prestigious position for reasons independent of their productivity at time 1. To the extent that the eminence of a scientist's mentor and the prestige of his doctoral department, *independently* of the productivity of the scientist, are particularistic criteria for evaluation, a *particularistic* advantage accumulates, not an advantage initiated for universalistic reasons. Rather than scientists who have earned their rewards at time 1 having an advantage at time 2, scientists who have had an advantage at time 1 multiply this advantage at time 2. In short, the reinforcing process of accumulative advantage appears to begin with a position obtained in time 1 for particularistic rather than universalistic reasons. Even if success in later job mobility is based more on objective criteria of productivity, the initial academic appointment, which is independent of earlier productivity, has a major impact on later productivity and hence the prestige of the second institutional location. Initial academic appointment significantly affects the chances of a scientist to become a successful researcher. Academic departments may recruit on the basis of the prestige of the mentor and the

¹³ Preliminary analysis has been completed to determine the relationship between collaboration and the departmental effect. This was done by constructing productivity measures in which each publication or citation was divided by the number of authors on the paper. While the departmental effect was slightly attenuated in regressions using these adjusted measures, it remained strong and statistically significant.

Table A1. Algorithm for Creating Yearly Citation Measures from Seven Volumes of *Science Citation Index* *

Year of Citation Measure	SCI-61							SCI-64							SCI-66				
	55	56	57	58	59	60	61	59	60	61	62	63	64		62	63	64	65	66
1955	+							.							.				
1956	+	+						.							.				
1957	+	+	+					.							.				
1958		+	+	+				.							.				
1959			+	+	+			.							.				
1960				+	+	+		.							.				
1961					+	+	+	.	+	+	+				.				
1962						+	+		+	+	+				.				
1963										+	+	+			.				
1964	.	.	.	.	.	.	.	.			+	+	+		.	+	+	+	
1965															.		+	+	+
1966															.			+	+
1967																			
1968																			
1969																			
1970																			
1971																			
1972																			
1973																			
1974																			

doctoral department because they have insufficient evidence of the young scientist's productivity. But nonetheless, this initial decision to hire, based on where one studied and with whom, has a major effect on the career of the scientist.

APPENDIX

The Construction of Yearly Citation Measures

Longitudinal analysis of the scientific career requires that yearly (or at least periodic) measures of productivity be available. For publication measures this is straightforward: each year of *Chemical Abstracts* (or some appropriate abstracting service) is searched for publications by the scientists under study; alternatively, the source index of *Science Citation Index* (SCI) can be used for the years after 1960. Yearly citation measures pose a greater problem since citation coding is costly and SCI was only first published in 1961. This appendix presents the procedures we have used to deal with these problems in constructing yearly citation measures.

Citations to each article written by members of our sample, whether they were the first author of the article or not, were coded from seven volumes of SCI: 1961, 1964, 1966, 1968, 1970, 1972, and 1974.

From this information citation measures were constructed to reflect the citations received by articles published in a given three-year period. Since the same article can be cited in many years, a decision had to be made regarding which citing years to use for constructing the measures. Table A1 depicts the scheme used. Vertically the year for which a citation count was to be constructed is listed. Horizontally are listed the volumes of SCI which were coded; associated with each volume of SCI is a listing of years in which the majority of articles cited in that volume had been published. In order to make the citation measures sensitive to changes in the rate in which an author's new work is being cited, each yearly count was restricted to publications for a three-year string of articles. A plus (+) in the body of Table A1 indicates that citations to articles from the appropriate year (as indicated by the years at the top or bottom of the table) out of the appropriate volume of SCI were summed to construct the yearly citation measure. In certain cases (e.g., the constructed count for 1964) citations from two volumes of SCI were used. This was done to take into account the half-life phenomenon in citations to articles: articles require time after publication before they can be cited; then, on the average, the number of citations increases before beginning to decline. In biochemistry substantial citation begins one to two years after publication. For example, for the year 1964 a measure of citations to articles published in 1962-1964

Table A1.—Continued

SCI-68					SC-70					SCI-72					SCI-74					Year of Citation Measure
64	65	66	67	68	66	67	68	69	70	68	69	70	71	72	70	71	72	73	74	
																				1955
																				1956
																				1957
																				1958
																				1959
																				1960
																				1961
																				1962
																				1963
																				1964
																				1965
.	+	+	+		.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	1966
.		+	+	+																1967
.			+	+	+	.	+	+	+											1968
.						.		+	+	+										1969
.						.		+	+	+	.	+	+	+						1970
.						.					.	+	+	+	.	.	.	.	.	1971
.						.					.		+	+	+	+	+	+		1972
.						.					.					+	+	+		1973
.						.					.						+	+	+	1974

VOLUME OF SCIENCE CITATION INDEX AND YEAR OF ARTICLE CITED

* Note: Each cell represents citations in a given volume of SCI to articles published in a given year. +’s indicate citations in that cell are used in three-year counts.

was desired. Since SCI volume 1964 was compiled too close to the year 1964 to have many citations to articles from that year, and since SCI volume 1966 was far enough away from articles published in 1962 that citations had begun to fall off, citations from both the 1964 and 1966 volumes of SCI were combined.

Since our sample members received their doctorates in a number of years, it was necessary to standardize these constructed counts by the year of the sample members’ Ph.D. Thus, for example, counts from all sample members who received their doctorates in 1957 were standardized to have a mean of zero and a variance of one. There were two major reasons why this standardization was necessary. First the coverage of SCI increases substantially from 1961 to 1974. For example, four years into their careers scientists receiving their doctorates in 1961 would have citation counts from a volume of SCI which surveyed approximately one and one-half times more journals than the corresponding volume of SCI for those receiving their degrees in 1957. It could thus appear that the 1962 cohort produced more cited articles than the 1957 cohort simply because a larger set of articles was surveyed for SCI.

Second, the average number of publications produced by biochemists increases rapidly at the start of

their careers. Due to the half-life of citations, being a year closer to the volume(s) of SCI used to construct the citation counts could make an appreciable difference in the magnitude of the citation measure. To avoid the possibility of giving a particular cohort of scientists an advantage in receiving citations simply because of the temporal relationship of their doctoral year to the years of SCI we used, we standardized the counts by year of the doctorate as described above.

After standardization the scores were transformed to approximate the scale of the raw counts. This was done by first multiplying the standardized scores by the standard deviation of the original raw counts, thereby changing the scale of the measures back to that of the raw counts. Then a constant was added so that the minimum value of the measures was equal to zero. In this way the measures could be treated as though they were raw, yearly counts.

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